



In-season effects of winter cover crops and tillage on soil NO₃ and N leaching – Analysis with generalized additive models

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ABSTRACT

Nitrate leaching during winter is a pressing environmental issue for agricultural systems in temperate climates. Interactions between cover crops and conservation tillage could have important implications for NO₃ leaching, both temporally in-season and aggregated per season. To investigate this, we carried out direct analysis of a subsoil NO₃ N concentration (nit-N) time-series using generalized additive models (GAMs) as well as generalized linear mixed effects modeling of aggregated seasonal N fluxes (N-leaching) derived using simulated subsoil water percolation. Subsoil NO₃ concentrations were measured for four seasons (2019–20, 2020–21, 2021–22 and 2022–23) in a field experiment located in Foulumgård, Denmark, involving three tillage treatments (ploughing, P; harrowing, H; and no-till; NT), three winter cover crop treatments (fodder radish as cover crop, Ref; weeds and volunteers, V; and bare soil during winter, B) and two crop rotations based on spring and winter cereals. GAM analysis showed that both V and Ref reduced subsoil nit-N relative to B, where the effect was greatest in Ref. No consistent effect was found for tillage treatments singly or in interaction with plant cover or different seasons. Differences in the timing or peak nit-N were solely determined by the season, likely due to different weather patterns. Ref reduced N-leaching by over 45 kg N ha⁻¹ y⁻¹ relative to B in average across seasons regardless of tillage, while V reduced it by 13–17 kg N ha⁻¹ y⁻¹ in tillage treatments P and H. In NT, however, there was no difference in either nit-N or N-leaching between V and B, suggesting weed growth and N uptake was suppressed after 20 years of consistent conservation tillage. Finally, we argue that direct analysis of subsoil NO₃ concentrations using GAMs represents a valuable complement to derived aggregated NO₃ leaching, which can provide useful insight about in-season temporal dimension of management practices and mitigation strategies.

1. Introduction

Nitrogen leaching is a major source of environmental nutrient pollution associated primarily with intensive fertilization in modern agricultural systems (Wang et al., 2019), as well as an important source of secondary GHG emissions (Pedersen and Olesen, 2002). In arable agricultural soils in temperate climates, a majority of all leached N takes the form of highly mobile NO₃ which washes out in autumn and winter when percolation is high and plant growth is slow (Di & Cameron, 2002). Since N is an essential plant macronutrient, reducing field N applications as a strategy to reduce NO₃ leaching and other reactive N emissions is limited by trade-offs in productivity and crop health, which can affect the economic sustainability of intensive agricultural systems

(Zhang et al., 2015). Therefore, mitigating NO₃ leaching from agriculture requires alternative agricultural practices that integrate multiple strategies and enhance the retention and cycling of nitrogen within the soil, increasing resource use efficiency at the field level without negatively affecting crop health and productivity.

Strategies to mitigate NO₃ leaching in arable agriculture (e.g., the methods reviewed by Cameron et al. (2013) generally involve adjusting the amount and timing of N fertilization to match the nutrient needs of the plants, fertilizing outside periods of high water percolation, boosting crop growth to increase their N uptake (e.g., by adjusting cultivars and protecting from diseases and pests), supplementing plant N uptake with ancillary plant cover (e.g., winter cover crops), and minimizing the presence of mobile NO₃ in soil via reduced or timely tillage and

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application of nitrification inhibitors, among others. Of these methods, winter cover crops and conservation tillage (reduced or no mechanical soil disturbance beyond sowing, avoidance of soil inversion) are of particular interest due to potential synergies and trade-offs between them, and because both form part of a number of emerging sustainable agricultural systems such as Conservation Agriculture (Hobbs et al., 2008), Regenerative Agriculture (Newton et al., 2020) and carbon farming practices (Bumbiere et al., 2022).

Winter cover crops, sometimes called catch crops, are widespread and well-studied in temperate climates (Aronsson et al., 2016) and can significantly reduce N leaching in arable soil by incorporating soil-mobile N into their biomass post-harvest (Abdalla et al., 2019; Thapa et al., 2018; Valkama et al., 2015). This N is later re-mobilized after the plant is destroyed either by frost or management. Here, depending on the weather, timing of destruction, and subsequent management, re-mobilized N can contribute to the nutrient supply of the succeeding crop, even potentially substituting some of the N fertilization in the following season (Vogeler et al., 2023). Conservation tillage, on the other hand, has been documented to affect soil organic carbon dynamics (Gómez-Muñoz et al., 2021) and improve soil physical and structural properties under certain conditions (Blanco-Canqui and Ruis, 2018; Mondal and Chakraborty, 2022), but its effect on NO₃ leaching appears to be highly dependent on soil type and concurrent management practices (Soane et al., 2012). While some research suggests conservation tillage can help reduce NO₃ leaching in agriculture by reducing SOM mineralization (Celik et al., 2017; Nila Rekha et al., 2011), a global meta-analysis of experimental results (Li et al., 2023) points towards greater leaching under conservation tillage due to increase total water drainage, but only in short-duration no-till and for certain soil texture and SOC ranges. Thus, the direct and indirect effects of conservation tillage on NO₃ leaching are complex and still not fully understood, particularly in the medium and long terms. Finally, conservation tillage regimes can affect crop root growth (Ruis and Blanco-Canqui, 2024) as well as the extent, location and timing of cover crop growth and later residue incorporation, and the local hydraulic properties and solute transport in the topsoil, particularly involving macropore flow (Cullum, 2009; Miranda-Vélez et al., 2022). Therefore, there remains a pressing necessity to further investigate the interactive effects of winter cover crops and conservation tillage on N leaching.

A common approach in field trials studying N-leaching is the repeated point determination of NO₃ concentrations in the subsoil using ceramic suction cups, which has a number of advantages over other soil N sampling methods (Wolf et al., 2023). Suction cups are relatively cheap and compact, can be installed in large numbers throughout an experimental field with relatively little soil disturbance, can remain installed for long periods of time, and can produce time-series of subsoil NO₃ concentrations covering a significant portion of the year, as long as there is sufficient moisture in the subsoil (Weihermüller et al., 2007). Thus, suction cup samples can capture valuable information regarding the temporal dimension of soil N surpluses and leaching losses. On the other hand, suction cup samples can be unreliable in structured soils with large amounts of macropore flow (Wang et al., 2012), cannot be taken continuously without risking disturbances to the soil hydrology around the sampling point (Weihermüller et al., 2007), and, importantly, do not provide direct measurements of either N or water fluxes.

Direct analysis and interpretation of NO₃ concentration time-series from suction cup samples is possible using traditional statistical tools (ANOVA, linear models, etc.), and has been used in the past in different forms. Some studies, for instance, have carried out independent comparisons of soil NO₃ concentrations segregated by sampling time (e.g., Macdonald et al., 2005), while others have compared average seasonal NO₃ concentrations (e.g., Hess et al., 2020; Premrov et al., 2014). These methods, however, can be impractical in large datasets covering long periods of time and/or when in-season temporal dynamics are of interest. Furthermore, yearly or seasonal N fluxes, not soil NO₃ concentrations, are typically the focus of monitoring and policy efforts.

Therefore, aggregated subsoil N fluxes, i.e., total N-leaching, are often derived from suction cup NO₃ concentration measurements using water fluxes measured elsewhere, estimated from soil water content or modelled from weather data using process-based models (e.g., Hansen and Djurhuus, 1997). Methods for the derivation of total N-leaching from NO₃ concentration time series typically involve of trapezoidal approximations of the integral of NO₃ over accumulated percolation (Lord and Shepherd, 1993), or step-wise approximations of daily NO₃ fluxes using either linear or water flux-weighted interpolation between consecutive suction cup samples (Vogeler et al., 2020). These calculations provide aggregated seasonal or yearly N-leaching values, which can be analyzed with relative ease using traditional statistical tools, such as linear or linear mixed models (Wey et al., 2021).

While the derivation and analysis of total N-leaching creates a useful summary figure of leaching emissions that is easy to communicate and interpret, it does not directly involve a quantitative assessment of the temporal variation of NO₃ concentrations and N fluxes. This leaves a knowledge gap regarding the time-dependent in-season effects of experimental treatments on subsoil NO₃ observations, and the derived or inferred N-leaching. This gap, however, can potentially be closed with the use of statistical tools better suited for the analysis of time-series data, such as generalized additive models (GAMs). GAMs are particularly well suited for analyzing nonlinear time-series data such as subsoil NO₃ concentrations, where the aim is to estimate the forms of the trends from the data rather imposing specific functional forms a priori. GAMs are both flexible and robust, being able to model nonlinear trends of varying shapes for multiple continuous covariates, including nonlinear interaction terms, as well as parametric terms for experimental factors as in most linear models. GAMs make very few assumptions about the nature of the trends in the response, and beyond the conditional distribution assumptions, we only further assume that the trends are smooth. Additionally, GAMs avoid overfitting by including penalty terms for the wiggleness of all smooth model terms. GAMs have been used to analyze the in-season variation of greenhouse gas emissions in agriculture (Jensen et al., 2022) and NO₃ leaching losses at the catchment level (McDowell et al., 2021), but remain underutilized in experimental studies of N-leaching in agriculture.

The objective of this study is to quantify the single and combined effects of tillage, winter cover crops and differences in weather and management across years on subsoil NO₃ concentration time series as well as derived seasonal N leaching. To this end, we have collected data from the CENTS long-term crop rotation and tillage experiment in Foulumgård, Denmark, which has had consistent conservation tillage treatments since 2003 and different winter cover crop treatments since 2017. We hypothesized that the cover crop reduced NO₃ concentration and N leaching in all tillage levels, but that lowest leaching was achieved for a combination of minimal tillage and winter cover cropping. The rationale for this hypothesis was two-fold. Firstly, we expected reduced SOM mineralization in conservation tillage concomitantly to the effects of cover crops. This reasoning is also one of the basic hypotheses behind the establishment of the CENTS experiment, following the results of (Hansen and Djurhuus, 1997) and (Nyborg and Malhi, 1989), among others. Secondly, we expected the decomposition of cover crop residues left on the surface under conservation tillage to be slower than that of residues mixed into the topsoil by plowing. This, in turn, we expected would lead to better chances of mineralized cover crop residue N being immobilized by microbes in the topsoil. We apply generalized additive models (GAMs) to subsoil NO₃ concentration time-series to analyze the in-season temporal effects of tillage, winter cover crops sampling season and infer their connection to increased or decreased risk of N leaching throughout the season. We then compare the resulting inferences with a traditional analysis involving generalized linear mixed modelling (GLMMs) of total N-leaching derived from interpolated NO₃ concentration time-series. We further discuss the accordance/discrepancies and advantages/disadvantages between GAM time-series inferences and GLMM aggregated flux inferences, and the respective importance of

both temporal evolution of soil NO_3^- and accumulated leaching. Finally, we discuss the potential role of GAMs as either a complement or a substitute to computing aggregated N fluxes in the field of agricultural N emissions.

2. Methods

2.1. Field experiment

In this study, we employ nitrate-bound N concentrations in soil water at 1 m depth (nit-N as mg N L^{-1}) measured in the CENTS experiment in Foulumgård (56°30' N, 9°35' E, Fig. 1), Denmark, between 2019 and 2023. The CENTS long-term crop rotation and tillage experiment at Aarhus University, Department of Agroecology, was established in 2002. The soil at Foulumgård is a Luvic Umbrisol (WRB, 2022) and has a sandy loam texture (9 % clay, 13 % silt, 75 % sand and 3.1 % organic matter) (ten Damme et al., 2025). The climate is humid temperate with a yearly average temperature of 8.2 °C and relatively high rainfall (avg. 678 mm y^{-1}) concentrated particularly in the autumn and winter seasons (Jørgensen et al., 2024). The weather at Foulumgård during the four years covered in this study (2019–2023) is summarized in Fig. 2.

The CENTS experiment follows a split-plot design (Fig. 1) consisting of four blocks divided into four main plots, each following a different crop rotation since 2002. In this study, we consider two crop rotations (Rotation) in the experiment, named R4 and R5 internally (Table 1), both consisting of spring-sown cereals, winter-sown cereals and grain legumes, where straw is always retained on-field after harvest. N fertilization follows crop-specific fertilizer quotas, corrections for residual effects of cover crops and yearly precipitation adjustments by soil type as specified in contemporary Danish regulations, resulting in a range of yearly N application rates throughout the experiment (Table 1). The main plots, in turn, are each divided into four plots that have consistently received different tillage treatments (Tillage) since 2002. Three Tillage treatments are considered here, consisting of inversion tillage to 0.2 m depth using a moldboard plough (P), harrowing to 0.08–0.1 m depth (H) and direct sowing with a chisel coulter-drill (no-till, NT). Finally, the tillage treatment plots are further divided into sub-plots. Beginning in 2017 in Foulumgård, three contrasting winter plant cover treatments (Cover) were established in different sub-plots in years preceding a spring-sown crop. The Cover treatments consist of a reference plot (Ref) where fodder radish (*Raphanus sativus* var. *oleiformis*) is sown as a cover crop shortly before or after harvest, weeds/volunteers

(V) where no management is carried out after harvest, and bare soil (B) between harvest and spring achieved by applying a broad-spectrum herbicide. The Ref treatment represents proactive management aimed at reducing NO_3^- leaching during autumn and winter and adheres to Denmark's compulsory cover crop schemes (Sommer and Knudsen, 2021). Weed and volunteer growth is not actively suppressed in Ref and is only potentially affected by the growth of the fodder radish cover crops. The V treatment represents land not intentionally sown with autumn-winter plant cover but not actively kept bare, as is common in land without compulsory cover crops due to current restrictions in tillage and herbicide application post-harvest. The B treatment represents an extreme practice where all plant cover post-harvest is chemically suppressed and serves principally as an experimental control for the V and Ref treatments. Since 2019, all sub-plots in the Cover treatment have been equipped with ceramic suction cups buried at 1 m depth for extracting soil water samples and quantifying nit-N. Two suction cups were installed in each sub-plot, which were pooled into a composite soil water sample at each sampling. Sampling took place at 2–3-week intervals during the autumn, winter and early spring seasons, resulting in a set of semi-sporadic nit-N time-series largely grouped by season (starting in June). In this design, treatment replication is fully nested within the block and treatment structure. Further details about the CENTS experiment can be found in Hansen et al. (2015).

2.2. Generalized additive models

Generalized additive models (GAMs) are a powerful and flexible statistical modelling tool that is used to estimate smooth functional relationships between a response variable and one or more continuous predictors (Pedersen et al., 2019). GAMs allow for the decomposition of the response variable into a set of smooth time-series functions corresponding to the different predictor variables. This, in turn, makes inferences regarding the temporal effect of different experimental treatments and covariates on the response variable possible. In the context of soil NO_3^- time-series, GAMs allow for the quantitative estimation of, and differentiation between, the effects of conservation tillage and of cover crops, as well as the extrinsic effects associated with different autumn-winter seasons (performance of the preceding cash crop, weather, unaccounted variations in plant growth, etc.) on soil NO_3^- over time.

GAMs are an extension of Generalized Linear Models (GLMs), where one or more effects of covariates on the response are assumed to be

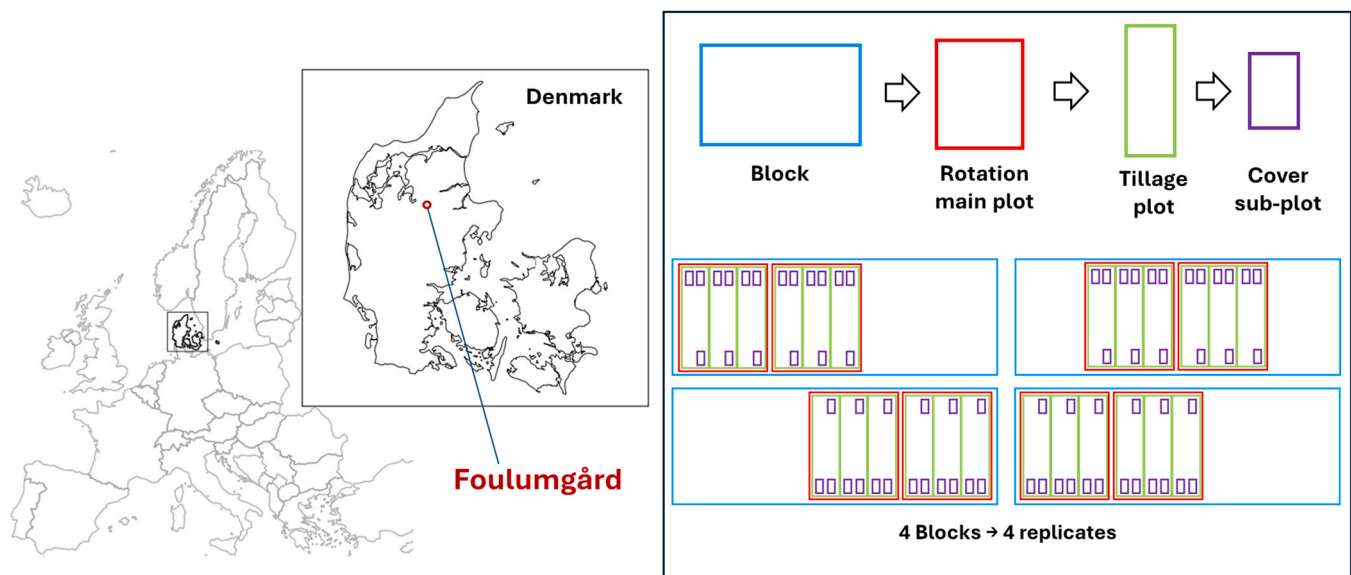


Fig. 1. Location of the CENTS LTE at Foulumgård, Denmark.

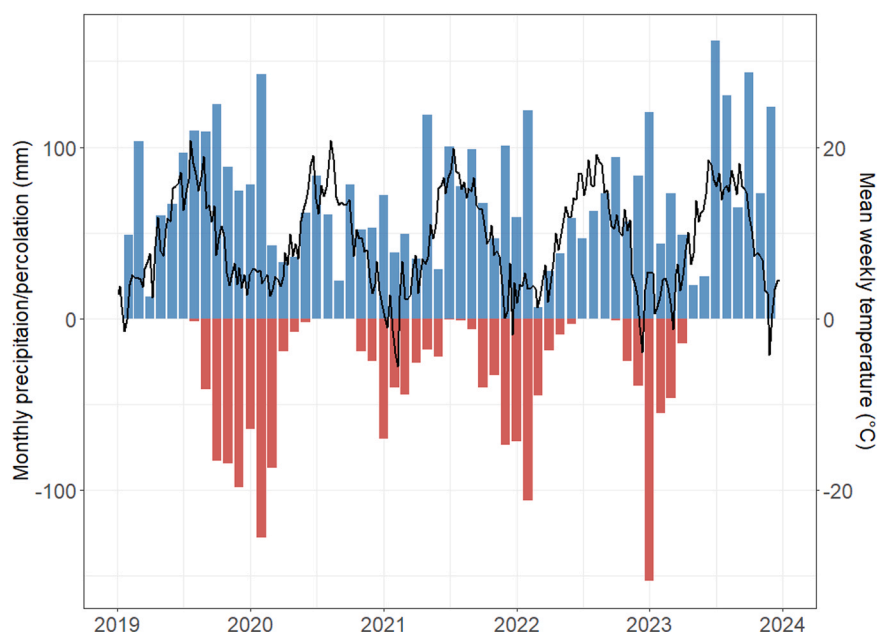


Fig. 2. Observed mean weekly temperature (line), observed monthly precipitation (blue bars) and simulated monthly percolation (red bars) in the CENTS experiment at Foulumgård, Denmark.

Table 1

Main crops and N fertilization rates in rotations R4 and R5 in the CENTS experiment at Foulumgård, listed by year of harvest.

Harvest year	R4				R5			
	Main crop	N rate (kg N ha ⁻¹)	Sowing date	Harvest date	Main crop	N rate (kg N ha ⁻¹)	Sowing date	Harvest date
2019	Oats (<i>Avena sativa</i>)	89	April 4th	August 19th	Spring Barley (<i>Hordeum vulgare</i>)	118	April 12th	August 12th
2020	Faba beans (<i>Vicia faba</i>)	0	March 31st	Sept. 1st	Oats (<i>Avena sativa</i>)	104	April 14th	August 17th
2021	Winter wheat (<i>Triticum aestivum</i>)	141	Sept. 30th (2020)	August 24th	Winter wheat (<i>Triticum aestivum</i>)	164	Sept. 30th (2020)	August 24th
2022	Spring Barley (<i>Hordeum vulgare</i>)	122	April 12th	August 24th	Spring barley (<i>Hordeum vulgare</i>)	122	April 25th	August 24th
2023	Oats (<i>Avena sativa</i>)	94	April 5th	August 15th	Spring barley (<i>Hordeum vulgare</i>)	122	April 4th	August 11th

smooth functions, the shapes of which are learned from the data themselves. Multiple smooths can be fitted simultaneously to describe the relationship between the dependent variable and a combination of continuous covariates and categorical factors. This approach is quite flexible, allowing the estimation of both smooth responses (e.g., as a function of time, temperature, etc.) and fixed factor effects affecting the centering of smooth curves.

In the present study, GAMs have been constructed with constrained factor smooth interactions in R v. 4.4.2 (R Core Team, 2024) using the “mgcv” package v. 1.9–1 (Wood, 2011; Wood et al., 2016). In the constrained factor structure, the model decomposes each group’s time series into a general or average smooth effect of time for the entire time-series data, plus group-specific difference smooths that capture the degree to which each group deviates from the average smooth. The group-specific smooths are parameterized to include the mean of the response for each group. The major benefit of the constrained factor smooth representation is that the resulting basis for the group-specific difference smooths is orthogonal to the average smooth of time, which makes the individual smooths identifiable. Higher order constrained factor smooth interactions are readily handled in the same way.

Concretely, we included in the model an average main-effect smooth describing the continuous change of soil nit-N concentrations throughout a year-long leaching sampling season starting in mid-June,

divided continuously into 52 season weeks (*Sweek*). We then included constrained factor smooths for each level of Rotation (R4, R5), Tillage (P, H, NT), Cover (Ref, V, B) and Season (2019–20, 2020–21, 2021–22 and 2022–23) representing the first-order interactive effects of the treatments. Additionally, we defined constrained factor smooths involving all possible second, third and fourth-order interactions involving the Rotation, Tillage, Cover and Season factors to account for their combined temporal effects. The relevance of higher-order interaction terms for the model fit was evaluated by comparing deviance and Aikake information criterion (Sakamoto et al., 1986) values between full and reduced nested models. The conditional distribution of the response was assumed to be distributed Tweedie (Dunn and Smyth, 2018) with a natural logarithm link function and automatic selection of the power index parameter (*p*). A small number of measurements (5 in total across all treatments) which fell below the limit of detection during periods of very low nit-N concentrations in the subsoil were assigned zero values. This allowed us to avoid fitting a considerably more complex censored model.

We calculated the partial effects of the model, i.e., the individual first, second, third and fourth-order interaction effects between *Sweek* and each level of the Cover, Tillage, Rotation and Season factors, respectively conditional upon all other model terms being set to zero. These partial effects, together with Bayesian 95 % credible intervals,

were calculated on the link scale using the R package “gratia” v. 0.9.0 (Simpson, 2024) and backtransformed to the response scale. Finally, we calculated nit-N estimates averaged over Season and Rotation conditional upon random effects being set to zero. This allowed us to infer the in-season temporal effects of Tillage and Cover on nit-N in an average plot and season. Conditional estimates and 95 % Bayesian credible intervals were calculated on the response scale and visualized using the R package “marginaleffects” (Arel-Bundock et al., 2024).

2.3. Field data interpolation and seasonal N-fluxes

In order to estimate aggregate leaching fluxes of nitrate-associated N, i.e. N-leaching, the sporadic nit-N time series were transformed into gapless daily time series via percolation-weighted interpolation as described and discussed in detail by Vogeler et al. (2020). Briefly, soil nit-N concentration at 1 m depth is assumed to only change as a direct function of water percolation. Therefore, the interpolation between two consecutive measurements is weighed according to modelled daily percolation (in mm d⁻¹) at 1 m depth which, in turn, is typically computed by a process-based model from daily weather data, modelled evapotranspiration and modelled hydrology. For each day (*i*) in an interpolation period (i.e., between consecutive sampling dates *a* and *b*) and daily modelled percolation value (*D*), the interpolated nit-N concentration (*C_i*) is equal to the fraction of the period's total percolation accumulated up to that day (*f_{D,i}*), multiplied by the difference in nit-N concentration between the sampling dates (*C_a* - *C_b*), following the equations:

$$C_i = C_a + (C_b - C_a)f_{D,i} \quad (1)$$

$$f_{D,i} = \frac{0.5D_a + \sum_{j=a+1}^{i-1} D_j + 0.5D_i}{0.5D_a + \sum_{j=a+1}^{b-1} D_j + 0.5D_b} \quad (2)$$

In this study, we employed Daisy, a flexible and comprehensive agro-hydrological model (Hansen, 2002) to estimate percolation. Using daily local weather data over the period 2005–2023, we constructed simulations of the hydric balance at field level for R4 and R5 crop rotations in the CENTS experiment. Precipitation data was corrected for wind speed effects following the method proposed by Allerup et al. (2000). Crop water abstraction in Daisy is determined by crop-specific plant models, with newly calibrated models for spring barley, winter wheat and fodder radish (Schwartzkopff et al., 2025, in review). Surface evapotranspiration was calculated by Daisy following the Penman-Monteith equation. Finally, Daisy uses the van Genuchten-Mualem model (Mualem, 1986; van Genuchten, 1980) for water retention and the Richards equation (Richards, 1931) for unsaturated water flow. The soil parametrization was carried out by adapting values previously developed for Foulumgård (see online documentation) with soil profile information reported by Schjøning (1992). No differences in the parametrization of the upper 20 cm of the soil column were implemented between Tillage treatments due to a lack of data suitable for evaluation.

Having computed a gapless daily nit-N concentration time-series, daily N-leaching fluxes were calculated as the product of daily nit-N concentration and modelled daily percolation and transformed into units of kg N ha⁻¹ d⁻¹. Finally, daily N fluxes were aggregated by sampling season (i.e. a calendar year starting on June 19th) into total-N (kg N ha⁻¹ y⁻¹) by sampling season.

2.4. Generalized linear mixed models

We analyzed the effects of crop rotation, tillage, winter plant cover and season on aggregated seasonal N leaching fluxes using a generalized linear mixed model with Template Model Builder, available in the package *glmmTMB* v. 1.1.9 (Brooks et al., 2017) in R v. 4.4.2 (Team, 2024). The response variable was aggregated seasonal N leaching derived from daily interpolated leaching values. As model predictors we

used crop rotation (*Rotation*), tillage (*Tillage*), winter plant cover (*Cover*) and sampling season (*Season*) along with all two-way and three-way interactions. Random effects were defined as experimental Block nested within the fourth-order interaction of Rotation, Tillage, Cover and Season, and the conditional distribution of the response, as with the GAM model, was assumed to be Tweedie with automatic selection of the power index parameter.

Finally, we calculated conditional model estimates of N-leaching in an average plot (i.e., random effects equal to zero) for different levels of either Cover, Tillage and Season, respectively grouped by one other main effect and averaged over the remaining main effect and Rotation. At each step, we computed pairwise comparisons of the N-leaching estimates with Tukey's family-wise p-value adjustments. Conditional estimates and pairwise comparisons were calculated using the *emmeans* package v. 1.10.1 (Lenth, 2024).

2.5. Scripts and data availability

A public GitHub repository containing the R script for data handling, analysis and visualization, as well as all input data and Daisy soil parametrization is available at: https://github.com/JorgeMirandaVelez/Miranda-Velez_2025_1

3. Results

The suction cup measurements show a strong variation in nit-N throughout each autumn-winter season, as well as between winter seasons and experimental treatments (Fig. 3). Most recognizable, perhaps, is the effect of Ref, where nit-N decreases during autumn, reaching a minimum of nearly zero in late December and early January, then increases again during spring. The exception is the winter season 2020–21, where autumn-sown winter wheat was grown in both R4 and R5 instead of winter cover crops. Here, nit-N increases during autumn and early winter, peaking in mid-January. On the other hand, nit-N at 1 m depth generally follows a similar trajectory in V and B across seasons, including 2020–21. In both V and B treatments, nit-N increases steadily during autumn and early winter, peaks between early and mid-winter, and decreases again towards spring. The timing of highest nit-N, and of highest potential N-leaching, varies between seasons as well, being noticeably early in season 2019–20. Interestingly, there is no obvious effect of the tillage treatments, temporal or overall.

3.1. Nit-N time-series with GAMs

Modelling with the chosen GAM produced a good fit to suction cup data and exhibited no obvious bias (Fig. 3). The adjusted R² and deviance explained for the model were 0.839 and 87.5 %, respectively. The fitted Tweedie *p* parameter was *p* = 1.285. Figs. 4 and 5 show, respectively, the partial effects of all first and second-order interactions, effectively representing a decomposition of nit-N among an underlying function of time and the time-dependent effects of individual experimental treatments and interactions throughout the season. Bayesian credible intervals (95 %) help visualize whether the different treatments and interactions have meaningful time-dependent effects on nit-N.

3.2. Tillage and cover crop smooth effects

The effects of winter plant cover alone on nit-N are clear. The partial effects of Cover show that Ref reduces nit-N markedly in autumn and early winter, reaching a minimum between weeks 20–25 of the sampling season (corresponding to early November to mid-December). Both the volunteer and bare soil treatments had the opposite effect, leading to increased nit-N towards the middle of the season, although the increase was greatest in B throughout the season.

The effect of Tillage alone is less straightforward than that of Cover. The partial effects of different Tillage levels are somewhat differentiated

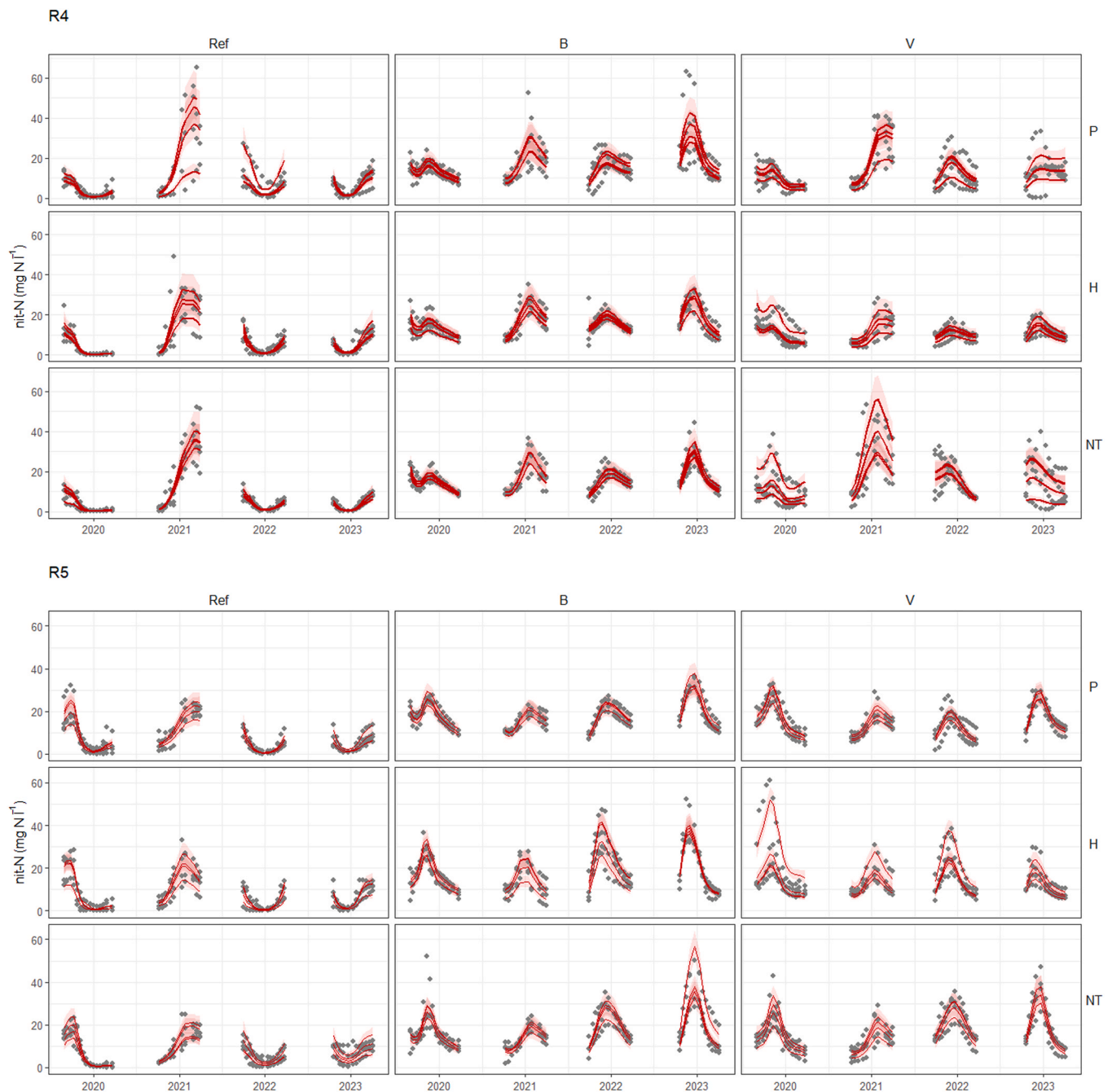


Fig. 3. Subsoil nitrate-bound N (nit-N) concentrations at 1 m depth in the CENTS experiment rotations R4 (top) and R5 (bottom). Columns correspond to winter plant cover treatments: fodder radish as reference winter cover crop (Ref), bare soil (B) and volunteers/weeds (V). Rows correspond to tillage treatments: inversion tillage with a moldboard plough (P), cultivation by harrowing at 8–10 cm depth (H) and no-till direct sowing with a chisel coulters (NT). Measurements took place between autumn and spring throughout four sampling seasons (2019–20, 2020–21, 2021–22 and 2022–23). Gray points represent suction cup measurements, while the red lines represent GAM estimates for each treatment replicate and shaded ribbons represent approximate 95 % Bayesian credible intervals.

throughout the season, but these differences are small relative to the overall effect magnitudes, and rarely larger than the 95 % credible interval. It is possible, however, that the different tillage treatments differentially affect the establishment and growth of winter cover crops and self-sown weeds and volunteers. If so, this effect would not be explained by Tillage alone but would be observable as a second-order interaction involving both Cover and Tillage.

The partial effects of Season, which lump together the preceding crop's N uptake, weather and other external factors, are well differentiated. This indicates that, independently from tillage or plant cover, external and unintentional differences between years have

unambiguous effects on subsoil nit-N throughout different winter seasons. Importantly, the sampling seasons of 2019–20 and 2020–21 both show periods of increased nit-N at different times. In 2019–20, there is a partial effect of increased nit-N early in the leaching season, corresponding to late summer and early autumn. This coincides with an unusually wet late summer and autumn in 2019 (Fig. 2), which could be expected to have caused an earlier onset of autumn percolation and N-leaching. The late winter increase in nit-N during the 2020–21 season would suggest slower NO_3^- transport in the soil profile, resulting in delayed breakthrough of nit-N. Indeed, a particularly dry summer in 2020 could have led to overall lower and later percolation in the

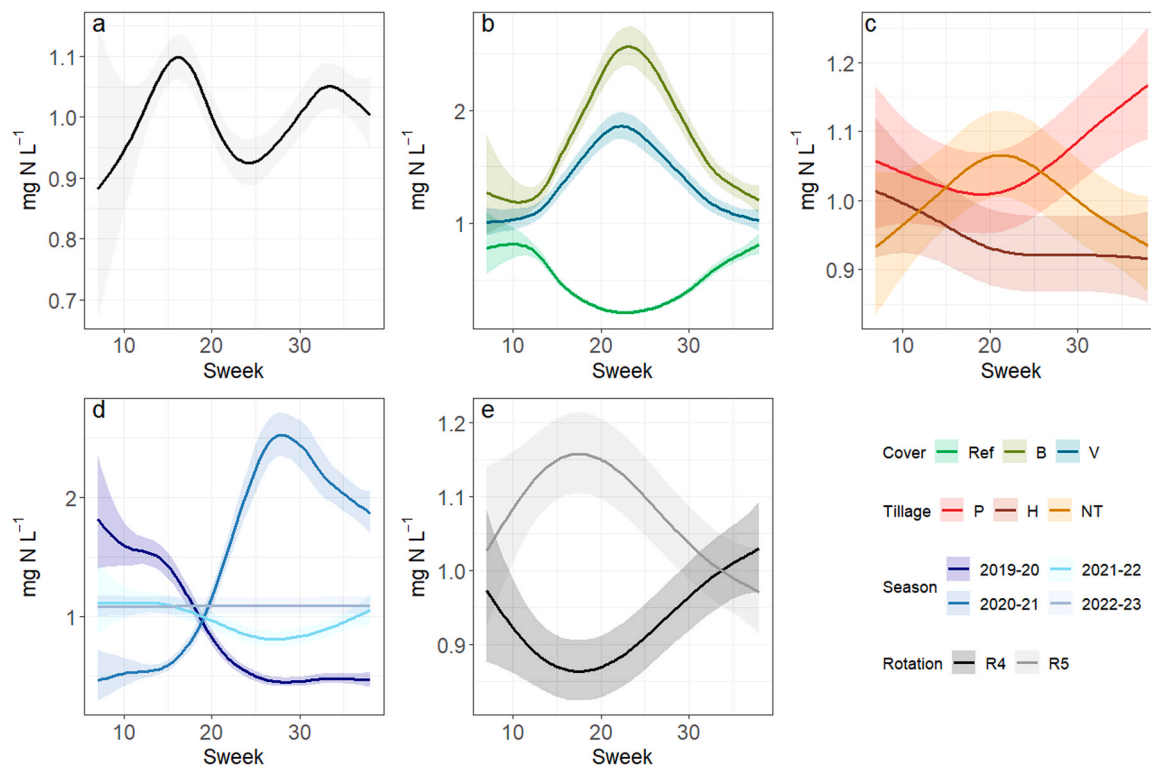


Fig. 4. GAM partial effect plots for subsoil nitrate-bound N concentrations (nit-N) as a function of time in a sampling season (weeks since mid-June, *Week*). The GAM was structured using constrained factor smooths, involving a mean smooth across treatments (a) and interaction terms between *Week* and plant cover (b), tillage (c), sampling season (d) and crop rotation (e). The shaded ribbons indicate Bayesian 95 % credible intervals. Note that the y axis is in the response scale of the model (mg N L^{-1}).

2020–21 season (Fig. 2), in turn delaying the leaching of nit-N. Thus, it is likely that Season alone reflects primarily differences in weather, which affected all plant cover and tillage treatments equally.

The partial effects of crop rotation alone were also clear, where nit-N was significantly increased by R5 throughout most of the leaching season relative to R4. This reflects the large effect that the main crops and corresponding N fertilization rates have in determining autumn and winter N leaching. Thus, the difference captured by the crop rotation reflects the generally lower N fertilization rates in R4 in 2019, 2021 and 2022, including the lack of N fertilization for faba beans in R4 2020.

3.3. GAM interactions involving plant cover, tillage, crop rotation and season

The second-order interactions between season week, plant cover and tillage (Fig. 5a) suggest a small trend of increased nit-N throughout the autumn-winter season in P relative to NT when reference cover crops are present. Conversely, there appears to be a small trend of increased nit-N in NT relative to both H and P throughout the season in the presence of weeds and volunteers. However, the small magnitude of effect differences relative to the error associated with the smooth components (and therefore the calculated confidence intervals) means none of these differences can be considered quantitatively different from underlying variation. Similarly, the interactions between season week, season and tillage (Fig. 5b) suggest small differences in the effect of tillage in different seasons, but in all cases these differences are small relative to the effects themselves and their associated error, and show no consistent patterns.

The interaction terms between season week, season and plant cover (Fig. 5c) show several season-dependent differences between cover crop treatments. Ref has a small but clear effect reducing nit-N relative to B and V in the second half of 2019–20 and mid-season in both 2021–22 and 2022–23. Also in 2022–23, there is an increase in nit-N relative to

both V and B towards the end of the season. A much larger interaction effect, however, is appreciable in season 2020–21, where Ref has a large increasing effect on nit-N, greatest in mid-winter, compared to B and V. Notably, these season-dependent differences between plant cover treatments are large relative to the corresponding model errors. Although the differences between cover crop treatments in 2019–20, 2021–22 and 2022–23 are difficult to interpret, the large increase in Ref in 2020–21 clearly reflects that winter wheat is significantly less effective at reducing subsoil nit-N than the winter cover crops grown in the Ref treatment in other years.

Interestingly, the interaction terms between season week, crop rotation and sampling season (Fig. 5d) show clear and opposing differences between R4 and R5 in 2019–20 and 2020–21. Here, nit-N is clearly higher in R5 relative to R4 throughout 2019–20, but clearly lower in R5 for most of 2020–21. This suggests a consistent effect by oats as main crop, where subsoil NO_3^- concentrations across winter cover crop and tillage treatments were lower following oats compared to the other crop in the same year. Importantly, this effect is present both in 2019–20, where the oats crop in R4 received 29 kg N ha^{-1} less than the spring barley crop in R5, and in 2020–21 where the oats in R5 received 104 kg N ha^{-1} and faba bean in R4 received no N fertilization. This difference, importantly, is not present in 2021–22 or 2022–23, where both rotations had winter wheat and spring barley, respectively, as main crop. Additionally, it is important to note that, although consistent and considerably larger than the model error, these differences are small relative to the partial effect sizes. On the other hand, neither the interaction between season week, crop rotation and tillage (Fig. 5e) nor between season week, crop rotation and plant cover (Fig. 5f) show clear differences in the effect of either tillage or plant cover on nit-N between rotations R4 and R5.

Finally, no consistent effects were found in the partial effects of third or fourth-order interaction terms (partial effect plots included in the [supplementary materials](#)). However, an analysis of deviance between

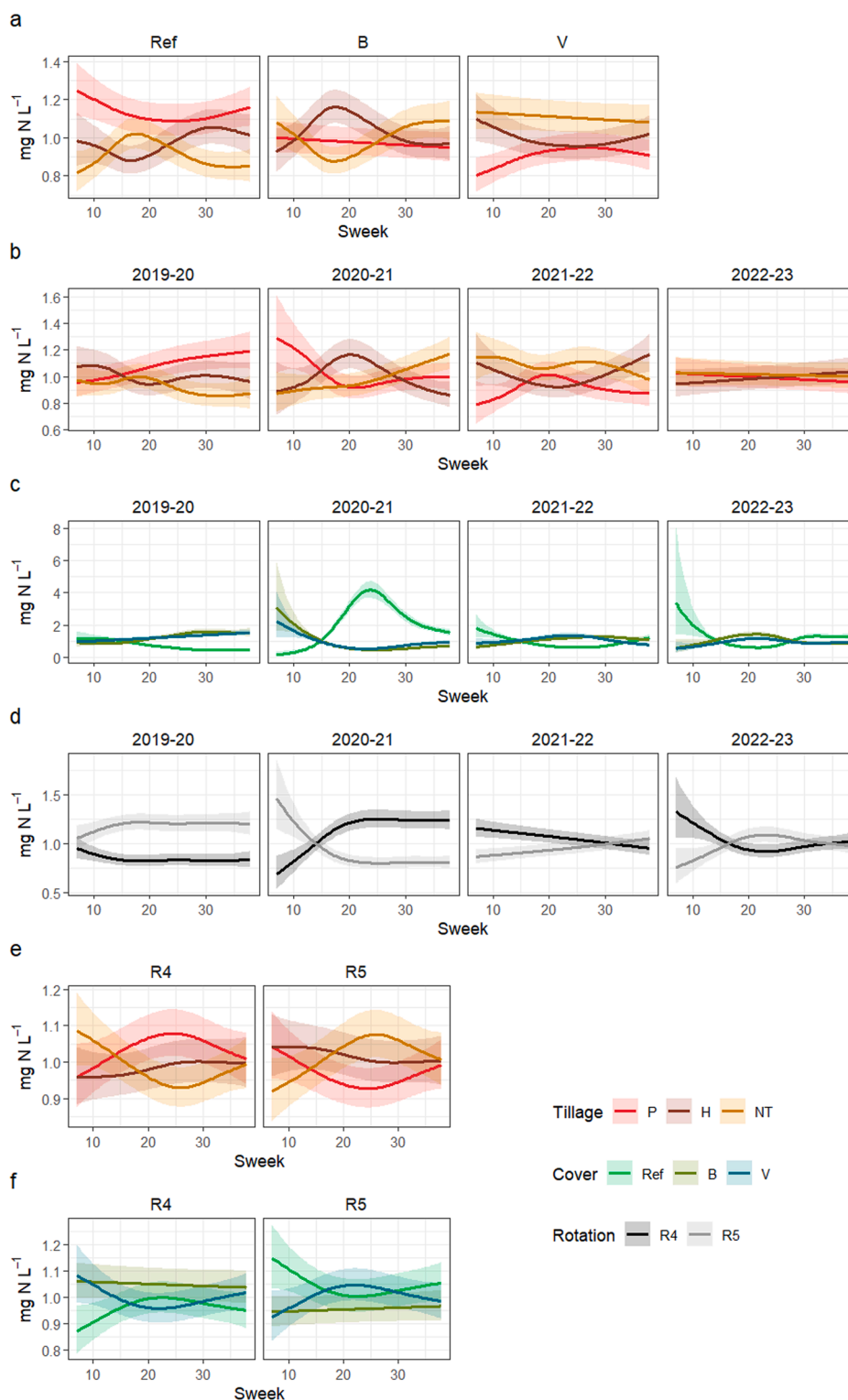


Fig. 5. GAM partial effects plots for subsoil nitrate-bound N concentrations (nit-N) as a function of time in a sampling season (weeks since mid-June, *Sweek*) and all second-order interactions with (a) winter plant cover and tillage, (b) sampling season and tillage, (c) sampling season and plant cover, (d) sampling season and crop rotation, (e) crop rotation and tillage and (f) crop rotation and plant cover. The shaded areas indicate Bayesian 95 % credible intervals. Note that the y axis is in the response scale of the model (mg N L^{-1}).

the full model and a reduced model with only first and second order interaction terms interactions yielded a deviance value difference of 223.8 in favor of the full model. Similarly, an Aikake information criterion comparison yielded an AIC difference of 319, in favor of the full model. This indicates that even though they do not present consistent

differences, third and fourth-order interaction terms are necessary in the present analysis.

3.4. Conditional smooth estimates for Cover and Tillage

Smooth estimates of nit-N over the winter season in different tillage and winter plant cover treatments and averaged across seasons and crop rotations (Fig. 6), illustrate the interaction between tillage and winter plant cover treatments. Comparisons grouped by winter plant cover show negligible differences among tillage treatments in both Ref and B. On the other hand, estimated nit-N in V was significantly higher in NT throughout the season relative to both H and P. This difference is considerable, up to 4.8 mg L^{-1} higher in NT compared to P in late November (Sweek=23), which would correspond roughly to 48 g N ha^{-1} additional N-leaching per mm of drainage during this period when weeds and volunteers are the only winter plant cover. Comparing between winter cover treatments grouped by tillage illustrates the same interaction weeds and volunteers and tillage. Here, we see a significant nit-N reduction in V relative to B, which is largest between November and December (Sweek 20–25), under both ploughed and harrowed treatments but not in no-till. This reduction, while much smaller than the difference between bare soil and reference cover crops, peaks in late November (Sweek 23) at 6.7 mg L^{-1} in P and 8.1 mg L^{-1} in H, corresponding roughly to 67 and 81 g N ha^{-1} lower N leaching per mm of drainage during this period.

3.5. Aggregated seasonal N-fluxes with GLMMs

Modelling of aggregated NO_3 leaching at 1 m depth for each sampling season (total N-leaching) with generalized linear mixed models showed overall significant effects by Rotation ($\chi^2=106.20$, $\text{df}=1$, $p < 0.001$), Tillage ($\chi^2=11.61$, $\text{df}=2$, $p = 0.003$), Cover ($\chi^2=642.55$, $\text{df}=2$, $p < 0.001$) and Season ($\chi^2=70.18$, $\text{df}=3$, $p < 0.001$) model predictors. Additionally, we found significant interactions between Tillage and Cover ($\chi^2=10.00$, $\text{df}=4$, $p = 0.040$), Rotation and Season

($\chi^2=89.00$, $\text{df}=3$, $p < 0.001$) and Cover and Season ($\chi^2=452.73$, $\text{df}=6$, $p < 0.001$) but not between Tillage and Season. Finally, we found significant three-way interactions between Rotation, Tillage and Season ($\chi^2=20.77$, $\text{df}=6$, $p = 0.002$), as well as between Rotation, Cover and Season ($\chi^2=17.79$, $\text{df}=6$, $p = 0.007$).

Conditional estimates of seasonal N fluxes at 1 m depth (N-leaching) segregated among Tillage levels and grouped by Cover (Table 2) show significantly higher N-leaching in NT compared to P and H under V, but no significant differences between tillage treatments in either Ref or B. Conversely, when segregating by Cover and grouping by Tillage, conditional estimates show significantly lower N-leaching in Ref compared to B by over $44 \text{ N ha}^{-1} \text{ y}^{-1}$ in all tillage treatments. In turn, N-leaching was significantly lower in V compared to B by $13.3 \text{ N ha}^{-1} \text{ y}^{-1}$ and $16.4 \text{ N ha}^{-1} \text{ y}^{-1}$ in P and H, respectively. In NT, however, there was no significant difference in N-leaching between V and B.

Conditional estimates segregated among Cover and Season and averaged over all other treatments (Table 3) show that N-leaching was significantly higher in 2019–20 compared to 2021–22 and 2022–23 across all Cover treatments and was significantly higher than all other seasons in V and B. Interestingly, N fertilization was lower in 2019 compared to 2021 and 2022 for both R4 and R5. This suggests that the high summer precipitation in the summer of 2019 led to increased total N-leaching across the experiment, regardless of winter plant cover and N fertilization rates. Unfortunately, crop-specific effects cannot be fully ruled out, as the main crops were different in 2019, 2020 and 2021 for both R4 and R5. Additionally, it is important to note that N-leaching was much lower in Ref compared to V and B for the same 2019–20 season. On the other hand, N-leaching was higher by over $30 \text{ kg N ha}^{-1} \text{ y}^{-1}$ in 2020–21 compared to all other seasons in Ref, while not significantly different between 2020 and 21, 2021–22 and 2022–23 in V and over $27 \text{ kg N ha}^{-1} \text{ y}^{-1}$ lower in 2020–21 compared to 2021–22 and 2022–23 in B. This reflects the substitution of winter plant cover with winter

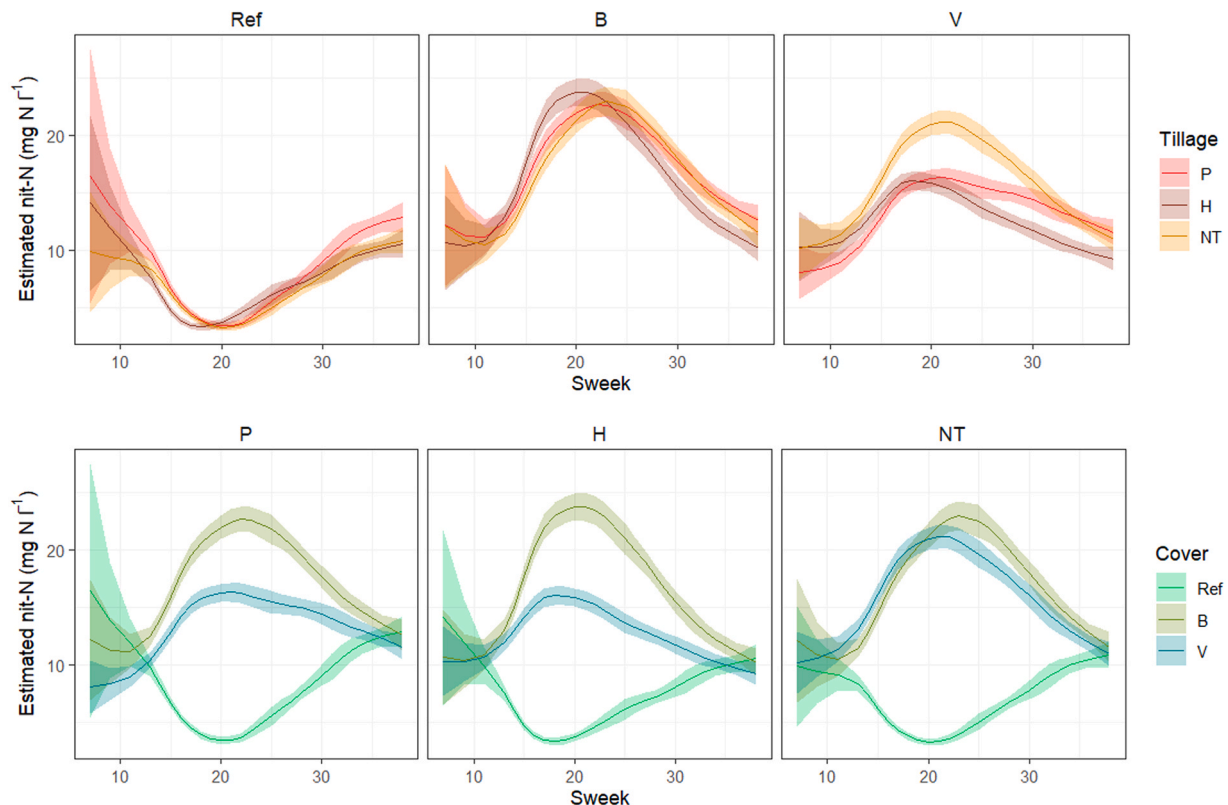


Fig. 6. Conditional GAM estimates of nit-N as a function of time (Sweek, weeks since mid-June) for ploughed (P), harrowed (H) and no-till (NT) tillage treatments, as well as reference winter cover crops (Ref), weeds and volunteers (V) and bare soil (B) as winter plant cover. The estimates are averaged over four winter seasons and two crop rotations. The shaded areas indicate Bayesian 95 % credible intervals. Note that the y axis is in the response scale of the model (mg N L^{-1}).

Table 2

Conditional estimates of seasonal N fluxes at 1 m depth (N-leaching) by winter plant cover (*Cover*) and tillage intensity (*Tillage*) treatments averaged over all other treatments. Grouping by Cover shows pairwise comparisons among Tillage levels within each level of Cover only, and vice versa for Cover grouped by Tillage. Comparisons were made in the model's link scale (natural logarithm). Different letters indicate significant differences ($p < 0.05$) between estimates.

Cover	Tillage	Estimated N-leaching	SE	
		kg N ha ⁻¹ y ⁻¹		
Ref	P	24.5	1.35	a
	H	21.1	1.21	a
	NT	21.1	1.24	a
V	P	56.1	2.26	a
	H	50.4	2.12	a
	NT	64.0	2.47	b
B	P	69.4	2.60	a
	H	66.8	2.54	a
	NT	70.0	2.62	a
Tillage P	Cover Ref	24.5	1.35	a
	V	56.1	2.26	b
	B	69.4	2.60	c
H	Ref	21.1	1.21	a
	V	50.4	2.12	b
	B	66.8	2.54	c
NT	Ref	21.1	1.24	a
	V	64.0	2.47	b
	B	70.0	2.62	b

wheat across Cover treatments in the autumn of 2020.

4. Discussion

4.1. Effects of season

Examination of the GAM results showed that it was largely external factors embedded in Season which controlled the timing of nit-N increases and decreases at 1 m depth, with no noticeable leading or lagging effects by tillage. Similarly, total N-leaching results indicate that high summer and autumn rainfall in 2019 led to greater total N-leaching within all winter cover treatments, although this increase was smaller than both the reduction in N-leaching by fodder radish cover crops

across Tillage treatments and the increase in leaching in 2020–21 caused by substitution of all plant cover treatments with winter wheat. Together, analyses of nit-N concentrations with GAMs and derived seasonal N-leaching suggest that weather conditions in 2019 led to earlier onset of NO₃ leaching and generally increased leaching compared to the other years in this study. However, it is important to underscore that while the Season was the primary factor determining differences in timing of nit-N increases and declines, winter plant cover had by far the greatest influence in both peak subsoil NO₃ concentrations and total N-leaching accumulated per season. Additionally, while crop-specific differences in N-fertilization and N-uptake represent potential confounding factors, it is important to note that all fertilization rates in this study adhere to the Danish N-fertilization norms, which are continuously adjusted to fit each crop's economic optimum yield response, slightly below the physiological yield response (Dalgaard et al., 2014; Sommer and Knudsen, 2021). Thus, differences in the efficiency of fertilizer N use between different crops have been reduced as much as possible. This supports previous findings in the field in similar soil and climate conditions that showed no significant effects of no-till on topsoil water infiltration rates compared to conventional tillage (Abdollahi et al., 2014). However, it is not certain that these results can be generalized to other soil types, particularly more heavily structured soils where preferential flow pathways play a more prominent role in water infiltration and solute transport (Daryanto et al., 2017; Jarvis, 2007). Indeed, while examining a wider variety of soils and climates, Blanco-Canqui and Ruis (2018) have previously found that no-till increased water infiltration and water holding capacity, while having no consistent effects on saturated water conductivity, which could indicate an increased influence of macropore flow in unsaturated conditions in no-till relative to conventional tillage.

In both analyses presented here, the effects of main crop and crop management are embedded in the effects of Season alone as well as its interactions with other predictors. In the GAM analysis, this is shown by differences between the partial effect smooths of rotations R4 and R5, as well as the smooth interaction terms between Rotation and Season in 2019–20 and 2020–21, which match the differences in spring fertilization and main crops. Likewise, modeling of seasonal N-leaching showed significant effects by Rotation as well as two and three-way interactions involving Rotation and Season, reflecting the differences in main crop and fertilization rates between R4 and R5. Furthermore, there was a significant interaction between Season and Cover in 2019–20 in both GAM and total N-leaching analyses, where the switch from the regular winter plant cover treatment to autumn-sown winter wheat is captured in the form of a large peak exclusive to the partial effect smooth for Ref and significantly higher estimates of leaching compared to other years in

Table 3

Conditional estimates of seasonal N fluxes at 1 m depth (N-leaching) by winter plant cover (*Cover*) and sampling season (*Season*) treatments averaged over all other treatments. Grouping by Cover shows pairwise comparisons among seasons within each level of Cover only and vice versa. Comparisons were made in the model's link scale (natural logarithm). Different letters indicate significant differences ($p < 0.05$) between estimates.

Cover	Season	N-leaching	SE		Season	Cover	N-leaching	SE	
		kg N ha ⁻¹ y ⁻¹					kg N ha ⁻¹ y ⁻¹		
Ref	2019–20	20.7	1.37	b	2019–20	Ref	20.7	1.37	a
	2020–21	52.5	2.49	c		V	76.5	3.21	b
	2021–22	15.9	1.13	a		B	89.1	3.52	c
	2022–23	14.0	1.05	a					
V	2019–20	76.5	3.21	b	2020–21	Ref	52.5	2.49	a
	2020–21	47.0	2.31	a		V	47.0	2.31	a
	2021–22	55.4	2.58	a		B	45.5	2.25	a
	2022–23	51.4	2.48	a	2021–22	Ref	15.9	1.13	a
B	2019–20	89.1	3.52	c		V	55.4	2.58	b
	2020–21	45.5	2.25	a		B	72.5	3.07	c
	2021–22	72.5	3.07	b	2022–23	Ref	14.0	1.05	a
	2022–23	75.9	3.18	b		V	51.4	2.48	b
						B	75.9	3.18	c

Ref, respectively. Assuming roughly equal water infiltration between Ref, B and V in 2020–21, this supports earlier research in similar soil and climate conditions showing that timely-sown winter wheat leads to significantly increased N-leaching compared to winter cover crops (Munkholm et al., 2017; Vogeler et al., 2021).

4.2. Effects of winter plant cover and tillage

As hypothesized, the total N-leaching results presented here add to the abundant experimental evidence that winter cover crops, in particular fodder radish, greatly reduce overall NO₃ leaching from cereal cropping systems in temperate climates compared to bare fallow, self-sown plant cover and winter cereals (De Notaris et al. 2018; Thapa et al. 2018; Tonitto et al. 2006; Valkama et al. 2015). Analysis with GAMs supports these results as well, as both model smooth terms and smooth conditional estimates show strong decreases in subsoil nit-N in Ref independently of tillage treatments, as well as more modest, but still significant reductions in nit-N with weeds and volunteers in P and H tillage treatments. Quantifying N-leaching reductions, Kumar et al. (2023) found reductions of up to 87 % with fodder radish sown in early August compared to bare soil in Foulumgård, while Aronsson et al. (2016) found a 45 % average NO₃ leaching reduction from winter cover crops compared to other winter management practices in a range of field studies across southern Scandinavia and Finland. In the present study, derived total N-leaching showed a reduction of over 67 % with fodder radish as a winter cover crop compared to bare soil in average across all other factors, and a 61 % reduction with cover crops compared to weeds and volunteers. Similarly, GAM estimates of nit-N for different Cover treatments averaged across Season and Rotation showed maximum reductions of 84 % and 80 % with reference winter cover crops compared to bare soil and weeds and volunteers, respectively. In both cases, the maximum nit-N reductions were seen during the month of November, a period in which percolation is very likely to be high in temperate climates like that of Denmark.

Total N-leaching was not significantly affected by tillage intensity in Ref or B Cover treatments across crop rotations and sampling seasons. Similarly, the GAM analysis of nit-N showed negligible differences in the timing and magnitude of peak nit-N in both Ref and B, as shown in constrained factor terms and smooth estimates averaged over Season and Rotation. Thus, both analyses show that consistent conservation tillage (H or NT) initiated in 2002 in Foulumgård had no meaningful effect relative to ploughing on either nit-N or total N-leaching amounts when winter plant cover consisted of either cover crops or bare soil across four years of measurements. Although in contradiction with our hypothesis, this supports other findings in the CENTS experiment, where there was no significant effect by tillage intensity on NO₃ leaching during the first 7 years of consistent management with winter cover crops and all three tillage intensities (Hansen et al., 2015) or during a single year after 16 years of consistent tillage and cover crop management (Taghizadeh-Toosi et al., 2022). Likewise, these findings agree with a meta-analysis of 28 studies investigating NO₃ leaching reductions by non-leguminous winter cover crops from Europe, the United States and New Zealand, which found no significant difference in the effect of cover crops between conventional and conservation tillage (Thapa et al., 2018). On the other hand, in a meta-analysis of 41 studies in the same geographical regions, Nouri et al. (2022) found increased NO₃ leaching under cover crops in conservation tillage relative to ploughing, which they propose could be attributed to the development of preferential flow pathways in medium and heavy-textured soils under no-till. Indeed, while examining a wide variety of soils and climates, Blanco-Canqui and Ruis (2018) found that no-till increased water infiltration and water holding capacity, while having no consistent effects on saturated water conductivity, which could indicate an increased influence of macropore flow in unsaturated conditions in no-till relative to conventional tillage.

An earlier review by Soane et al. (2012) proposes that the difficulty in finding scientific consensus regarding the effects of tillage on soil NO₃

concentrations and NO₃ leaching is in part explained by the variability of water flow pathways in structured soils. This explanation implies that lightly textured soils do not develop strong preferential flow pathways under no-till. This implication is supported by previous field measurements in Foulumgård that showed no significant effects of no-till on topsoil water infiltration rates compared to conventional tillage (Abdollahi et al., 2014). Thus, the present results suggest that in lightly textured soils in temperate regions, conservation tillage has a minimal effect on both water infiltration and flow, soil NO₃ transport, as well as the capacity of winter cover crops to reduce NO₃ leaching. However, it is also likely that the lack of effects on nit-N and N-leaching by conservation tillage presented here cannot be generalized to more strongly structured soils where preferential flow pathways generally play a more prominent role in water infiltration and solute transport (Daryanto et al., 2017; Jarvis, 2007). Indeed, Li et al. (2023) found a clear influence of soil texture on the effect of no-till on NO₃ leaching, where no-till reduced area-scaled leaching in soils with fine texture by 34 % but had no effect in coarsely textured soils. This highlights the need for studies simultaneously investigating subsoil NO₃ and different water flow regimes under different tillage intensities for soils with contrasting textures.

In Cover treatment V, where winter plant cover consisted of self-sown volunteers and weeds, no-till management led to significantly higher N-leaching amounts compared to both P and H in average across crop rotations and sampling seasons. This interaction is also shown by the GAM analysis, where conditional estimates in V show higher nit-N in NT relative to both P and H throughout most of the sampling season. Thus, there is a very clear indication that NT hindered the capacity of weeds and volunteers to uptake N after harvest and reduce autumn NO₃ concentrations at 1 m depth. Two general hypothetical mechanisms could explain this effect: i) NT hinders the germination and establishment of grain spilled during harvest, and ii) NT hinders the germination and/or growth of weed seeds present in the soil's seed bank or has resulted in a reduced weed seed bank over time.

The first mechanism is not supported by the present results, as there were no significant differences in nit-N between tillage treatments in the fodder radish cover crop treatment. Since the fodder radish was established by broadcasting shortly before or after harvest, and thus encountered similar surface conditions as volunteer grain, this would suggest that NT did not generally hinder seed germination at the surface around harvest. It is, however, possible that fodder radish is better suited for broadcasting than the cereals and faba beans grown as main crops during this study. Indeed, some experimental evidence exists showing that cereals used as winter cover crops suffer worse establishment and N uptake rates when broadcast compared to drilled or cultivated (e.g., Fisher et al. 2011). However, this appears to be a poorly explored topic, as we were unable to find direct experimental comparisons of establishment and N uptake in broadcast cereal and mustard family winter cover crops in temperate climates.

The second possible mechanism, that NT hindered weed establishment in V, is supported by existing research showing that a majority of agricultural weeds in Denmark are seed-borne annual plants (Melander et al., 2013) and that no-till contributes to the depletion of annual weed seedbanks in the soil over time (Nichols et al., 2015). This depletion occurs because in the absence of inversion tillage, wild seeds are not buried in the soil where they are protected from predation and unfavorable weather (Chauhan et al., 2012; Kumar et al., 2020). At the same time, weed seeds accumulating near the surface are more likely to germinate and, in systems where this is allowed, trigger management with herbicides (Melander et al., 2013). However, it is important to note that Scherner et al. (2016) found increased incidence of the grass weeds *A. spica-venti* and *V. myuros* in H and NT throughout the first 11 years of consistent tillage management in the Flakkebjerg site of the CENTS long-term experiment, where *A. spica-venti* was introduced intentionally in 2002. Here, the authors argue that although weed seed losses are greatest at the surface, burial by inversion tillage would greatly hinder the emergence of these two grass weeds and could be used strategically

in Conservation Tillage systems to control similar weeds. Thus, although the effect may vary for certain types of weeds and no experimental confirmation is yet available, the most compelling explanation for the present results is that the weed seedbank is generally suppressed in NT after two decades of consistent no-till management and ad-hoc weed control using herbicides. In turn, this would have led to negligible N-uptake by weeds in V, and thus increased nit-N and total N-leaching when compared to V in P or H. In general, the present results show that, although significant, the capacity of weeds and volunteers to reduce NO₃ leaching relative to bare soil is small and is somehow further suppressed in the long-term by no-till.

Here, it is important to note that effective weed control in the cropping season, usually through chemical means, is key to Conservation Agriculture practices (Jug et al., 2025). Thus, if the observed effect of long-term no-till on autumn weeds and volunteers is indeed caused by a depletion of the soil seedbank, there is likely no alternative to establishing effective dedicated cover crops after harvest if the goal is to reduce N-leaching. Finally, it is unclear under which conditions and for how long the effects of no-till on weeds and volunteers would remain after a change in management. Thus, it could potentially lead to delayed colonization with wild species and prolonged periods of increased N-leaching following an end to cropping, such as in re-wilding or long-term set-aside efforts.

4.3. Discrepancies and synergies between methods

The two methods for analysis of suction cup NO₃ time series data used in this study have different benefits and downsides, depending on the research objectives and the assumptions made about the soil system that surround the data. Aggregated NO₃ leaching fluxes are more easily communicated and more directly related to N losses from agricultural soil and concomitant N loadings to the environment than estimates of soil NO₃ concentrations over time. Furthermore, concrete NO₃ annual or seasonal leaching figures are often the desired product of experimental studies, for instance, in the context of compiling emissions inventories as well as comparing and recommending mitigation strategies (e.g., Eriksson et al. 2020). However, calculating NO₃ leaching fluxes requires percolation data, which introduces a potential source of bias and/or error. Percolation is remarkably difficult to measure reliably in the field, as burying measuring equipment is likely to disturb water flow paths around the measuring point. Similarly, utilizing percolation data from a nearby lysimeter introduces potential errors caused by edge effects on the lysimeter walls, and biases caused by the constructed nature of the lysimeter soil column. In both cases, the presence and magnitude of measuring errors is unknown. This problem extends to the difficulties surrounding model evaluation, if a model is used for calculating percolation. As is the case here, adequate data for parametrizing differences in tillage treatments was not available, and thus N-leaching fluxes were calculated using a generic parametrization. Even when there is percolation or soil water content data available for model parametrization and evaluation, the potential for unknown biases and errors remains, including issues of equifinality. Lastly, many long-term experiments do not include equipment for measuring percolation at all. Thus, the unknown biases and errors related to deriving NO₃ leaching fluxes from suction cup data are particularly difficult to remove or even quantify. This underscores a first argument for directly analyzing the nit-N data before carrying out other calculations, as this is the only data where variation, error and bias associated with the experiment can be quantified.

A second argument for directly analyzing nit-N data arises when considering the effects of treatments on a slightly larger scale. If the patterns of water infiltration and movement can be assumed to be variable but broadly homogeneous at the field scale, across treatments, it is possible to infer general differences in potential NO₃ leaching from subsoil NO₃ concentrations with the help of NO₃ concentration estimates averaged across years or seasons. In well-structured soils, where

tillage intensity is more likely to affect water infiltration and solute movement, the key question becomes whether these effects are primarily restricted to the depth of soil affected by tillage. If so, increased percolation near the surface would accelerate the removal of NO₃ from the topsoil and its transport to the subsoil, but the volume of percolation and solute transport through the bulk of the soil profile would remain largely homogeneous across tillage treatments. Here, the inference of NO₃ leaching risks from NO₃ concentrations remains sensible. On the other hand, if one expects that increased preferential flow caused by conservation tillage affects solute transport throughout the soil profile, these inferences would be invalid. For instance, if no-till management in a strongly structured soil leads to the development of long connected macropores reaching below the root zone, significant amounts of NO₃ leaching could occur during periods of low precipitation, where large percolation volumes would not be expected. This, however, is a problem that equally affects the derivation of aggregated NO₃ leaching fluxes using simulated or estimated percolation and requires either direct observation of water flow regimes in the subsoil, or careful modelling of preferential flow, both of which are difficult tasks. With these considerations in mind, it can be argued that direct statistical analysis of NO₃ concentration time series with GAMs or similar tools can be as informative as deriving aggregated leaching fluxes, so long as concrete leaching emission figures are not required. Furthermore, direct analysis of NO₃ concentrations makes it possible to identify and isolate the temporal effects of management interventions on potential leaching and quantitatively explore their interactions with other time-dependent factors such as crop growth or weather patterns.

A recent study by Gould et al. (2025) in the fields of evolutionary and conservation ecology investigated the different conclusions reached by a set of researchers and research groups pursuing independent analyses of identical datasets. They reached the conclusion that different analysis methodologies of the same data can lead to meaningfully different conclusions being drawn, but that no one analysis method could be identified as unequivocally correct among the others. Their suggestion, rather, is that no one analysis option can be considered as complete or universally reliable, and that there is an advantage in exploring and reporting more than one analysis method given a set of experimental results. This principle arguably applies to the field of ecosystem ecology and agricultural ecology as well, particularly in the present context, where analysis of experimental data at different levels (directly or after derivation of summary quantities) involves intrinsically different analytical approaches. Thus, direct analysis with GAMs and subsequent derivation of aggregated leaching are perhaps best seen as complementary analyses, which can lend solidity to each other where they agree, and can inform any conclusions with crucial extra context where don't agree.

A final important consideration is the apparent difficulties in deriving p-values for hypothesis testing when implementing analyses with GAMs. Certain tests are quickly available to GAM users, such as approximated p-values for testing equality to zero of smooths (Wood, 2012), essentially testing whether a covariate has any effect on the predicted variable. Similarly, effect sizes of parametric (not smooth) terms can be tested in GAMs in the same way as in a GLM. Finally, although the estimates and contrasts derived from GAMs are not single values but functions over a continuous range, estimate comparisons and contrasts at fixed covariate values can be extracted from the model and used in hypothesis testing (i.e. determining significant differences based on p-values). However, there is real evidence in academic publishing of an over-reliance on p-values and statistical significance instead of effect sizes, uncertainty estimation and ultimately scientific relevance (Nakagawa and Cuthill, 2007; Sullivan and Feinn, 2012). Thus, rather than worrying about whether GAMs can be used to carry out familiar comparisons and tests, we would argue that direct analysis of time-series data with GAMs should be seen as a valuable opportunity to re-focus from watching significance thresholds to examining the temporality, magnitude and consistency of smooth effects. Thus, it is highly likely

that the analysis of complex non-linear relationships with GAMs can lead to more insightful explorations of soil NO₃ and similar experimental datasets.

5. Conclusions

The present study set out to investigate the single and combined effects of winter cover crops and conservation tillage on NO₃ leaching in two arable crop rotations in the same long-term conservation tillage experiment. We found that under plowing and harrowing management, NO₃ concentrations and total NO₃ leaching were slightly reduced by weeds and volunteers and greatly reduced by winter cover crops. This adds to the already considerable body of evidence documenting the effectiveness of winter cover crops to reduce NO₃ leaching in arable systems in temperate regions. Furthermore, we found that conservation tillage had no clear effect on subsoil NO₃ concentrations or seasonal NO₃ leaching, either individually or in interaction with cover crops. This refutes our expectation that conservation tillage would act synergistically with cover crops in reducing N-leaching, even after long-term conservation tillage. Lastly, weeds and volunteers under no-till management did not lead to any meaningful reductions in subsoil NO₃ concentrations or NO₃ leaching compared to bare soil, which we argue is likely caused by a depletion of the annual weed soil seedbank in the no-till plots after more than 20 years of no-till and ad-hoc weed control. However, direct in-situ evidence of soil bank depletion or reduced weed growth would be needed to confirm this.

Additionally, this study aimed to discuss the potential of direct analysis of soil NO₃ time series with GAMs to falsify, substitute or complement the more widely used methods that involve aggregating derived NO₃ N fluxes. Here, we find that, except for not being able to produce concrete leaching emission figures, direct analysis of NO₃ concentrations with GAMs is very informative for comparing potential N-leaching at different times and under different management practices. Thus, given the inherent difficulties in producing and validating percolation data, the direct analysis of NO₃ concentrations is a valuable complement to the derivation of aggregated N-leaching fluxes, leading to more complete and insightful analyses when both approaches are used in conjunction.

CRedit authorship contribution statement

Jorge Federico Miranda-Vélez: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gavin L. Simpson:** Writing – review & editing, Validation, Methodology, Formal analysis. **Elly Møller Hansen:** Writing – review & editing, Supervision, Data curation. **Lars J. Munkholm:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the

online version at doi:10.1016/j.agee.2025.110011.

Data availability

A link to an open GitHub repository containing all data was included in the manuscript

References

- Abdalla, M., Hastings, A., Cheng, K., Yue, Q., Chadwick, D., Espenberg, M., Truu, J., Rees, R.M., Smith, P., 2019. A critical review of the impacts of cover crops on nitrogen leaching, net greenhouse gas balance and crop productivity. *Glob. Chang. Biol.* 25 (8), 2530–2543. <https://doi.org/10.1111/gcb.14644>.
- Abdollahi, L., Munkholm, L.J., Garboud, A., 2014. Tillage system and cover crop effects on soil quality: II. Pore characteristics. *Soil Sci. Soc. Am. J.* 78 (1), 271–279. <https://doi.org/10.2136/sssaj2013.07.0302>.
- Allerup, P., Madsen, H., Vejen, F., 2000. Correction of precipitation based on off-site weather information. *Atmos. Res.* 53 (4), 231–250. [https://doi.org/10.1016/S0169-8095\(99\)00051-4](https://doi.org/10.1016/S0169-8095(99)00051-4).
- Arel-Bundock, V., Greifer, N., Heiss, A., 2024. How to interpret statistical models using marginal effects for r and python. *J. Stat. Softw.* 111 (9). <https://doi.org/10.18637/jss.v111.i09>.
- Aronsson, H., Hansen, E.M., Thomsen, I.K., Liu, J., Ogaard, A.F., Kankanen, H., Ulen, B., 2016. The ability of cover crops to reduce nitrogen and phosphorus losses from arable land in Southern scandinavia and Finland. *J. Soil Water Conserv.* 71 (1), 41–55. <https://doi.org/10.2489/jswc.71.1.41>.
- Blanco-Canqui, H., Ruis, S.J., 2018. No-tillage and soil physical environment. *Geoderma* 326, 164–200. <https://doi.org/10.1016/j.geoderma.2018.03.011>.
- Brooks, M.E., Kristensen, K., Benthem, K.J. v., Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Mächler, M., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for Zero-inflated generalized linear mixed modeling. *R. J.* 9 (2), 378–400. <https://doi.org/10.32614/RJ-2017-066>.
- Bumbiere, K., Diaz Sanchez, F.A., Pubule, J., Blumberga, D., 2022. Development and assessment of carbon farming solutions. *Environ. Clim. Technol.* 26 (1), 898–916. <https://doi.org/10.2478/rtuct-2022-0068>.
- Cameron, K.C., Di, H.J., Moir, J.L., 2013. Nitrogen losses from the soil/plant system: a review. *Ann. Appl. Biol.* 162 (2), 145–173. <https://doi.org/10.1111/aab.12014>.
- Celik, I., Günel, H., Acar, M., Gök, M., Bereket Barut, Z., Pamiralan, H., 2017. Long-term tillage and residue management effect on soil compaction and nitrate leaching in a typical haploxerert soil. *Int. J. Plant Prod.* 11 (1), 131–149. <https://doi.org/10.22069/ijpp.2017.3314>.
- Chauhan, B.S., Singh, R.G., Mahajan, G., 2012. Ecology and management of weeds under conservation agriculture: a review. *Crop Prot.* 38, 57–65. <https://doi.org/10.1016/j.cropro.2012.03.010>.
- Cullum, R.F., 2009. Macropore flow estimations under no-till and till systems. *Catena* 78 (1), 87–91. <https://doi.org/10.1016/j.catena.2009.03.004>.
- Dalgaard, T., Hansen, B., Hasler, B., Hertel, O., Hutchings, N.J., Jacobsen, B.H., Jensen, L.S., Kronvang, B., Olesen, J.E., Schjørring, J.K., Kristensen, I.S., Graversgaard, M., Termansen, M., Vejre, H., 2014. Policies for agricultural nitrogen management—trends, challenges and prospects for improved efficiency in Denmark. *Environ. Res. Lett.* 9, 115002. <https://doi.org/10.1088/1748-9326/9/11/115002>.
- ten Damme, L., Goberna, M., Sánchez-Moreno, S., Pulido-Moncada, M., Philippot, L., Ros, M., Bragazza, L., Hallin, S., Feiziane, D., Munkholm, L.J., 2025. Long-Term application of No-Tillage-Induced greater risk of poor topsoil aeration along a european pedoclimatic gradient. *Eur. J. Soil Sci.* 76 (2). <https://doi.org/10.1111/ejss.70081>.
- Daryanto, S., Wang, L., Jacinthe, P.A., 2017. Impacts of no-tillage management on nitrate loss from corn, soybean and wheat cultivation: a meta-analysis. *Sci. Rep.* 7 (1), 12117. <https://doi.org/10.1038/s41598-017-12383-7>.
- De Notaris, C., Rasmussen, J., Sørensen, P., Olesen, J.E., 2018. Nitrogen leaching: a crop rotation perspective on the effect of n surplus, field management and use of catch crops. *Agric. Ecosyst. Environ.* 255, 1–11. <https://doi.org/10.1016/j.agee.2017.12.009>.
- Di, H.J., Cameron, K.C., 2002. Nitrate leaching in temperate agroecosystems: sources, factors and mitigating strategies. *Nutr. Cycl. Agroecosystems* 64, 237–256. <https://doi.org/10.1023/A:1021471531188>.
- Dunn, P.K., Smyth, G.K., 2018. Generalized linear models with examples in r. Springer, New York, NY. <https://doi.org/10.1007/978-1-4419-0118-7>.
- Eriksen, J., Thomsen, I.K., Hoffmann, C.C., Hasler, B., & Jacobsen, B.H. (2020). VIRKEMIDLER TIL REDUKTION AF KVÆLSTOFBELASTNINGEN AF VANDMILJØET (DCA Rapport, Issue. D-N. C. f. f. o. Jordbrug. <https://dcapub.au.dk/djfpdf/DCArapport174.pdf>).
- Fisher, K.A., Momen, B., Kratochvil, R.J., 2011. Is broadcasting seed an effective winter cover crop planting method? *Agron. J.* 103 (2), 472–478. <https://doi.org/10.2134/agronj2010.0318>.
- van Genuchten, M.T., 1980. A Closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Sci. Soc. Am. J.* 44 (5), 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>.
- Gómez-Muñoz, B., Jensen, L.S., Munkholm, L., Olesen, J.E., Møller Hansen, E., Bruun, S., 2021. Long-term effect of tillage and straw retention in conservation agriculture systems on soil carbon storage. *Soil Sci. Soc. Am. J.* 85 (5), 1465–1478. <https://doi.org/10.1002/saj2.20312>.

- Gould, E., Fraser, H.S., Parker, T.H., Nakagawa, S., Griffith, S.C., Vesik, P.A., Fidler, F., Hamilton, D.G., Abbey-Lee, R.N., Abbott, J.K., Aguirre, L.A., Alcaraz, C., Aloni, I., Altschul, D., Arekar, K., Atkins, J.W., Atkinson, J., Baker, C.M., Barrett, M., Zitomer, R.A., 2025. Same data, different analysts: variation in effect sizes due to analytical decisions in ecology and evolutionary biology. *BMC Biol.* 23 (1). <https://doi.org/10.1186/s12915-024-02101-x>.
- Hansen, E.M., Djourhuus, J., 1997. Nitrate leaching as influenced by soil tillage and catch crop. *Soil Tillage Res.* 41 (3-4), 203–219. [https://doi.org/10.1016/S0167-1987\(96\)01097-5](https://doi.org/10.1016/S0167-1987(96)01097-5).
- Hansen, E.M., Munkholm, L.J., Olesen, J.E., Melander, B., 2015. Nitrate leaching, yields and carbon sequestration after noninversion tillage, catch crops, and straw retention. *J. Environ. Qual.* 44 (3), 868–881. <https://doi.org/10.2134/jeq2014.11.0482>.
- Hansen, S., 2002. Daisy, a flexible soil-plant-atmosphere system model. *Model. Doc.* 1–47.
- Hess, L.J.T., Hinckley, E.-L.S., Robertson, G.P., Matson, P.A., 2020. Rainfall intensification increases nitrate leaching from tilled but not no-till cropping systems in the U.S. Midwest. *Agric. Ecosyst. Environ.* 290. <https://doi.org/10.1016/j.agee.2019.106747>.
- Hobbs, P.R., Sayre, K., Gupta, R., 2008. The role of conservation agriculture in sustainable agriculture. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 363 (1491), 543–555. <https://doi.org/10.1098/rstb.2007.2169>.
- Jarvis, N.J., 2007. A review of non-equilibrium water flow and solute transport in soil macropores: principles, controlling factors and consequences for water quality. *Eur. J. Soil Sci.* 58 (3), 523–546. <https://doi.org/10.1111/j.1365-2389.2007.00915.x>.
- Jensen, S.A., Webb, J.R., Simpson, G.L., Baulch, H.M., Leavitt, P.R., Finlay, K., 2022. Seasonal variability of CO₂, CH₄, and N₂O content and fluxes in small agricultural reservoirs of the Northern great plains. *Front. Environ. Sci.* 10. <https://doi.org/10.3389/fenvs.2022.895531>.
- Jørgensen, M.S., Plauborg, F., & Kørup, K. (2024). *CLIMATE NORMAL FOR FOULUM 1991-2020* (223). (DCA Report, Issue. D. D. C. f. a. Agriculture. (<https://dcapub.au.dk/djfpublikation/djfpdf/DCArapport223.pdf>).
- Jug, D., Jug, I., Brozović, B., Šeremešić, S., Dolijanović, Ž., Zsembeli, J., Ujj, A., Marjanović, J., Smutny, V., Dušková, S., Neudert, L., Macák, M., Wilczewski, E., Đurđević, B., 2025. Conservation soil tillage: bridging science and farmer Expectations—An overview from Southern to Northern Europe. *Agriculture* 15 (3). <https://doi.org/10.3390/agriculture15030260>.
- Kumar, U., Thomsen, I.K., Eriksen, J., Vogeler, I., Mäenpää, M., Hansen, E.M., 2023. Delaying sowing of cover crops decreases the ability to reduce nitrate leaching. *Agric. Ecosyst. Environ.* 355. <https://doi.org/10.1016/j.agee.2023.108598>.
- Kumar, V., Mahajan, G., Dahiya, S., Chauhan, B.S., 2020. Challenges and opportunities for weed management in No-Till farming systems. In: Dang, In.Y.P., Dalal, R.C., Menzies, N.W. (Eds.), *No-till Farming Systems for Sustainable Agriculture: Challenges and Opportunities*. Springer International Publishing, pp. 107–125. https://doi.org/10.1007/978-3-030-46409-7_7.
- Lenth, R.V. (2024). emmeans: Estimated Marginal Means, aka Least-Squares Means. In: Li, J., Hu, W., Chau, H.W., Beare, M., Cichota, R., Teixeira, E., Moore, T., Di, H., Cameron, K., Guo, J., Xu, L., 2023. Response of nitrate leaching to no-tillage is dependent on soil, climate, and management factors: a global meta-analysis. *Glob. Chang. Biol.* 29 (8), 2172–2187. <https://doi.org/10.1111/gcb.16618>.
- Lord, E.I., Shepherd, M.A., 1993. Developments in the use of porous ceramic cups for measuring nitrate leaching. *J. Soil Sci.* 44 (3), 435–449. <https://doi.org/10.1111/j.1365-2389.1993.tb00466.x>.
- Macdonald, A.J., Poulton, P.R., Howe, M.T., Goulding, K.W.T., Powlson, D.S., 2005. The use of cover crops in cereal-based cropping systems to control nitrate leaching in SE England. *Plant Soil* 273 (1-2), 355–373. <https://doi.org/10.1007/s11040-005-0193-3>.
- McDowell, R.W., Simpson, Z.P., Ausseil, A.G., Etheridge, Z., Law, R., 2021. The implications of lag times between nitrate leaching losses and riverine loads for water quality policy. *Sci. Rep.* 11 (1), 16450. <https://doi.org/10.1038/s41598-021-95302-1>.
- Melander, B., Munier-Jolain, N., Charles, R., Wirth, J., Schwarz, J., van der Weide, R., Bonin, L., Jensen, P.K., Kudsk, P., 2013. European perspectives on the adoption of nonchemical weed management in Reduced-Tillage systems for arable crops. *Weed Technol.* 27 (1), 231–240. <https://doi.org/10.1614/WT-D-12-00066.1>.
- Miranda-Vélez, J.F., Diamantopoulos, E., Vogeler, I., 2022. Does macropore flow in no-till systems bypass mobile soil nitrogen after harvest? *Soil Tillage Res.* 221. <https://doi.org/10.1016/j.still.2022.105408>.
- Mondal, S., Chakraborty, D., 2022. Global meta-analysis suggests that no-tillage favourably changes soil structure and porosity. *Geoderma* 405. <https://doi.org/10.1016/j.geoderma.2021.115443>.
- Mualem, Y., 1986. Hydraulic conductivity of unsaturated soils: prediction and formulas. In: Klute, In.A. (Ed.), *Methods of Soil Analysis Part 1* (2nd ed.). American Society of Agronomy, Inc. Soil Science Society of America, Inc.
- Munkholm, L.J., Hansen, E.M., Thomsen, I.K., Wahlström, E.M., Østergaard, H.S., de Varennes, A., 2017. Nitrogen uptake, nitrate leaching and root development in winter-grown wheat and fodder radish. *Soil Use Manag.* 33 (2), 233–242. <https://doi.org/10.1111/sum.12337>.
- Nakagawa, S., Cuthill, I.C., 2007. Effect size, confidence interval and statistical significance: a practical guide for biologists. *Biol. Rev.* 82 (4), 591–605. <https://doi.org/10.1111/j.1469-185x.2007.00027.x>.
- Newton, P., Civita, N., Frankel-Goldwater, L., Bartel, K., Johns, C., 2020. What is regenerative agriculture? A review of scholar and practitioner definitions based on processes and outcomes. *Front. Sustain. Food Syst.* 4. <https://doi.org/10.3389/fsufs.2020.577723>.
- Nichols, V., Verhulst, N., Cox, R., Govaerts, B., 2015. Weed dynamics and conservation agriculture principles: a review. *Field Crops Res.* 183, 56–68. <https://doi.org/10.1016/j.fcr.2015.07.012>.
- Nila Rekha, P., Kanwar, R.S., Nayak, A.K., Hoang, C.K., Pederson, C.H., 2011. Nitrate leaching to shallow groundwater systems from agricultural fields with different management practices. *J. Environ. Monit.* 13 (9), 2550–2558. <https://doi.org/10.1039/c1em10120j>.
- Nouri, A., Lukas, S., Singh, S., Singh, S., Machado, S., 2022. When do cover crops reduce nitrate leaching? A global meta-analysis. *Glob. Chang. Biol.* 28 (15), 4736–4749. <https://doi.org/10.1111/gcb.16269>.
- Nyborg, M., Malhi, S.S., 1989. Effect of zero and conventional tillage on barley yield and nitrate nitrogen content, moisture and temperature of soil in north-central Alberta. *Soil Tillage Res.* 15 (1-2), 1–9. [https://doi.org/10.1016/0167-1987\(89\)90059-7](https://doi.org/10.1016/0167-1987(89)90059-7).
- Pedersen, E.J., Miller, D.L., Simpson, G.L., Ross, N., 2019. Hierarchical generalized additive models in ecology: an introduction with mgcv. *PeerJ* 7, e6876. <https://doi.org/10.7717/peerj.6876>.
- Pedersen, S.O., Olesen, J.E., 2002. Greenhouse gas inventories for agriculture in the nordic countries. *Greenh. Gas. Invent. Agric. Nord. Ctries. Helsingør.*
- Premrov, A., Coxon, C.E., Hackett, R., Kirwan, L., Richards, K.G., 2014. Effects of over-winter Green cover on soil solution nitrate concentrations beneath tillage land. *Sci. Total Environ.* 470–471, 967–974. <https://doi.org/10.1016/j.scitotenv.2013.10.057>.
- Richards, L.A., 1931. Capillary conduction of liquids through porous mediums. *Physics* 1 (5), 318–333. <https://doi.org/10.1063/1.1745010>.
- Ruis, S.J., Blanco-Canqui, H., 2024. How does no-till affect soil-profile distribution of roots? *Can. J. Soil Sci.* 104 (4), 350–361. <https://doi.org/10.1139/cjss-2023-0099>.
- Sakamoto, Y., Ishiguro, M., Kitagawa, G., 1986. Akaike information criterion statistics. *Dordr. Neth. D. Reidel* 81 (10.5555), 26853.
- Scherner, A., Melander, B., Kudsk, P., 2016. Vertical distribution and composition of weed seeds within the plough layer after eleven years of contrasting crop rotation and tillage schemes. *Soil Tillage Res.* 161, 135–142. <https://doi.org/10.1016/j.still.2016.04.005>.
- Schjønnung, P. (1992). *Karakterisering af jordarealer omkring Forskningscenter Foulum* (S 2229-1992). (Tidskrift for Planteavl's Specialserie, Issue.
- Schwartzkopf, M.V., Miranda-Vélez, J.F., Abrahamsen, P., Grønning, C.R., & Jensen, L. S. (2025). Multi crop-model calibration and evaluation in the agroecosystem model Daisy - a comparison of nitrogen leaching in cereal rotations with and without catch-crops (Under Review). *Nutrient Cycling in Agroecosystems*.
- Simpson, G.L., 2024. Gratia: an r package for exploring generalized additive models. *J. Open Source Softw.* 9 (104). <https://doi.org/10.21105/joss.06962>.
- Soane, B.D., Ball, B.C., Arvidsson, J., Basch, G., Moreno, F., Roger-Estrade, J., 2012. No-till in Northern, Western and south-Western Europe: a review of problems and opportunities for crop production and the environment. *Soil Tillage Res.* 118, 66–87. <https://doi.org/10.1016/j.still.2011.10.015>.
- Sommer, S.G., Knudsen, L., 2021. Impact of danish livestock and manure management regulations on nitrogen pollution, crop production, and economy. *Front. Sustain.* 2. <https://doi.org/10.3389/frsus.2021.658231>.
- Sullivan, G.M., Feinn, R., 2012. Using effect Size—or why the p value is not enough. *J. Grad. Med. Educ.* 4 (3), 279–282. <https://doi.org/10.4300/jgme-d-12-00156.1>.
- Taghizadeh-Toosi, A., Hansen, E.M., Olesen, J.E., Baral, K.R., Petersen, S.O., 2022. Interactive effects of straw management, tillage, and a cover crop on nitrous oxide emissions and nitrate leaching from a sandy loam soil. *Sci. Total Environ.* 828, 154316. <https://doi.org/10.1016/j.scitotenv.2022.154316>.
- Team, R.C., 2024. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.
- Thapa, R., Mirsky, S.B., Tully, K.L., 2018. Cover crops reduce nitrate leaching in agroecosystems: a global meta-analysis. *J. Environ. Qual.* 47 (6), 1400–1411. <https://doi.org/10.2134/jeq2018.03.0107>.
- Tonitto, C., David, M.B., Drinkwater, L.E., 2006. Replacing bare fallows with cover crops in fertilizer-intensive cropping systems: a meta-analysis of crop yield and n dynamics. *Agric. Ecosyst. Environ.* 112 (1), 58–72. <https://doi.org/10.1016/j.agee.2005.07.003>.
- Valkama, E., Lemola, R., Känkänen, H., Turtola, E., 2015. Meta-analysis of the effects of undersown catch crops on nitrogen leaching loss and grain yields in the nordic countries. *Agric. Ecosyst. Environ.* 203, 93–101. <https://doi.org/10.1016/j.agee.2015.01.023>.
- Vogeler, I., Hansen, E.M., Nielsen, S., Labouriau, R., Cichota, R., Olesen, J.E., Thomsen, I. K., 2020. Nitrate leaching from suction cup data: influence of method of drainage calculation and concentration interpolation. *J. Environ. Qual.* 49 (2), 440–449. <https://doi.org/10.1002/jeq2.20020>.
- Vogeler, I., Jensen, J.L., Thomsen, I.K., Labouriau, R., Hansen, E.M., 2021. Fertiliser n rates interact with sowing time and catch crops in cereals and affect yield and nitrate leaching. *Eur. J. Agron.* 124. <https://doi.org/10.1016/j.eja.2021.126244>.
- Vogeler, I., Hansen, E.M., Thomsen, I.K., 2023. The effect of catch crops in spring barley on nitrate leaching and their fertilizer replacement value. *Agric. Ecosyst. Environ.* 343. <https://doi.org/10.1016/j.agee.2022.108282>.
- Wang, Q., Cameron, K., Buchan, G., Zhao, L., Zhang, E.H., Smith, N., Carrick, S., 2012. Comparison of lysimeters and porous ceramic cups for measuring nitrate leaching in different soil types. *N. Z. J. Agric. Res.* 55 (4), 333–345. <https://doi.org/10.1080/00288233.2012.706224>.
- Wang, Y., Ying, H., Yin, Y., Zheng, H., Cui, Z., 2019. Estimating soil nitrate leaching of nitrogen fertilizer from global meta-analysis. *Sci. Total Environ.* 657, 96–102. <https://doi.org/10.1016/j.scitotenv.2018.12.029>.
- Weiermüller, L., Siemens, J., Deurer, M., Knoblauch, S., Rupp, H., Gottlein, A., Putz, T., 2007. In situ soil water extraction: a review. *J. Environ. Qual.* 36 (6), 1735–1748. <https://doi.org/10.2134/jeq2007.0218>.

- Wey, H., Hunkeler, D., Bischoff, W.A., Bunemann, E.K., 2021. Field-scale monitoring of nitrate leaching in agriculture: assessment of three methods. *Environ. Monit. Assess.* 194 (1), 4. <https://doi.org/10.1007/s10661-021-09605-x>.
- Wolf, K.A., Pullens, J.W.M., Borgesen, C.D., 2023. Optimized number of suction cups required to predict annual nitrate leaching under varying conditions in Denmark. *J. Environ. Manag.* 328, 116964. <https://doi.org/10.1016/j.jenvman.2022.116964>.
- Wood, S.N., 2011. Fast stable restricted maximum likelihood and marginal likelihood estimation of semiparametric generalized linear models. *J. R. Stat. Soc. Ser. B Stat. Methodol.* 73 (1), 3–36. <https://doi.org/10.1111/j.1467-9868.2010.00749.x>.
- Wood, S.N., 2012. On p-values for smooth components of an extended generalized additive model. *Biometrika* 100 (1), 221–228. <https://doi.org/10.1093/biomet/ass048>.
- Wood, S.N., Pya, N., Säfken, B., 2016. Smoothing parameter and model selection for general smooth models. *J. Am. Stat. Assoc.* 111 (516), 1548–1563. <https://doi.org/10.1080/01621459.2016.1180986>.
- WRB, I.W.G., 2022. World Reference Base for Soil Resources. International soil classification system for naming soils and creating legends for soil maps, 4th ed.). International Union of Soil Sciences (IUSS).
- Zhang, X., Davidson, E.A., Mauzerall, D.L., Searchinger, T.D., Dumas, P., Shen, Y., 2015. Managing nitrogen for sustainable development. *Nature* 528 (7580), 51–59. <https://doi.org/10.1038/nature15743>.